

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_failure_replay.
- Best TSPLIB holdout gap: tsplib_failure_replay.
- Best synthetic holdout gap: tsplib_failure_replay.

Run Metadata

- run_name: run_20260513_150328_o
- started_at_local: 2026-05-13 15:03:28
- finished_at_local: 2026-05-13 15:13:57
- duration_hhmm: 00:10
- duration_seconds: 629.322
- seed_offset: 14000
- replicate_label: o
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.116589	0.010343	0.077954	0.664145	0.76	0.034776
tsplib_random_replay	random	score_only	False	0.213387	0.064916	0.159398	0.793498	0.56	-0.031386
tsplib_failure_replay	failure	score_only	False	0.096403	0.0	0.061347	0.679064	0.76	0.028617
tsplib_failure_replay_compression	failure	novelty_gate	True	0.203768	0.0	0.129671	0.683973	0.76	0.020722

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.116589, synthetic 0.010343, combined 0.077954.

- Accepted-epoch count 3, mean accepted code novelty 0.664145, and final complexity 0.76.
- Adaptation efficiency 0.034776 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.116589 across 7 instances; family means: ch=0.126198, kroD=0.125059, pcb=0.201603, pr=0.041208, rd=0.065487, st=0.13037.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 18952, "family": "lin", "name": "lin105", "optimality_gap": 0.318033}, {"best_known_cost": 2579, "cost": 3385, "family": "a", "name": "a280", "optimality_gap": 0.312524}, {"best_known_cost": 7542, "cost": 9203, "family": "berlin", "name": "berlin52", "optimality_gap": 0.220233}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.793498, and final complexity 0.56.
- Adaptation efficiency -0.031386 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.096403, synthetic 0.0, combined 0.061347.
- Accepted-epoch count 3, mean accepted code novelty 0.679064, and final complexity 0.76.
- Adaptation efficiency 0.028617 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.096403 across 7 instances; family means: ch=0.086971, kroD=0.084014, pcb=0.25448, pr=0.119805, rd=0.020354, st=0.022222.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3371, "family": "a", "name": "a280", "optimality_gap": 0.307096}, {"best_known_cost": 14379, "cost": 18335, "family": "lin", "name": "lin105", "optimality_gap": 0.275123}, {"best_known_cost": 629, "cost": 763, "family": "eil", "name": "eil101", "optimality_gap": 0.213037}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.203768, synthetic 0.0, combined 0.129671.
- Accepted-epoch count 3, mean accepted code novelty 0.683973, and final complexity 0.76.

- Adaptation efficiency 0.020722 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.203768 across 7 instances; family means: ch=0.173538, kroD=0.233117, pcb=0.229568, pr=0.316423, rd=0.171302, st=0.128889.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3460, "family": "a", "name": "a280", "optimality_gap": 0.341605}, {"best_known_cost": 14379, "cost": 16420, "family": "lin", "name": "lin105", "optimality_gap": 0.141943}, {"best_known_cost": 7542, "cost": 8479, "family": "berlin", "name": "berlin52", "optimality_gap": 0.124238}]

Judge Appendix

Summary of Results: Replay-Aware TSP Benchmark Suite

Condition	Synthetic Holdout Gap	TSPLIB Holdout Gap	Transfer Gap	Replay Mode	Final Complexity	Code Novelty (Mean)	Notes
tsplib_failure_replay	0.000	0.0964	0.0613	Failure replay	0.76	0.679	Best performance for synthetic, TSPLIB holdout, and transfer gaps.
tsplib_no_replay	0.0103	0.1166	0.0780	None	0.76	0.664	Moderate gaps; no replay baseline.
tsplib_failure_replay_compression	0.000	0.2038	0.1297	Failure replay + compression	0.76	0.684	Transfer and TSPLIB gaps worse than failure replay without compression despite similar novelty.
tsplib_random_replay	0.0649	0.2134	0.1594	Random replay	0.56	0.793	Worst transfer and TSPLIB gaps despite highest code novelty; lower model complexity.

Interpretation

- **Best Transfer and Optimality Performance**

The **tsplib_failure_replay** condition yields the lowest holdout gaps on both TSPLIB (0.0964) and synthetic benchmarks (0.0), as well as the lowest transfer gap (0.0613). This indicates superior transfer capability and optimization quality under failure replay.

- **No Replay vs Failure Replay**

Compared to the **no replay** baseline, failure replay improves holdout gaps (TSPLIB gap reduced from 0.1166 to 0.0964; transfer gap from 0.0780 to 0.0613), supporting that failure replay is beneficial for generalization and transfer.

- **Random Replay**

Random replay leads to substantially worse transfer and holdout gaps (TSPLIB: 0.2134; transfer: 0.1594). Despite having the highest code novelty (0.793 mean) and lower complexity (0.56), random replay harms optimality and transfer, showing that lexical/code novelty alone is not indicative of better algorithms.

- **Failure Replay with Compression Pressure**

Although compression pressure is applied here, the **tsplib_failure_replay_compression** condition performs worse than failure replay alone, doubling the TSPLIB gap to 0.204 and transfer gap to 0.130. Code novelty remains similar to failure replay without compression, indicating compression reduces transfer efficacy without adding algorithmic innovation.

Conclusion

- **Failure Replay (tsplib_failure_replay)** is the most effective replay mode, yielding the best transfer and optimality-gap metrics, outperforming both no replay and random replay.
- **Code novelty does not correlate positively with transfer performance:** random replay shows higher novelty but worse gaps.
- **Compression-aware replay reduces transfer quality** compared to failure replay without compression, despite similar novelty and complexity.

The evidence stresses prioritizing **optimality-gap and transfer metrics over code novelty or speculative narrative**. Failure replay yields improved transfer and should be favored in replay-aware TSP optimization benchmarks.